

## Slope-intercept form



1  $b$  stands for  $y$ -intercept and  $m$  stands for slope.

2

a Slope =  $-\frac{4}{3}$ ,  $y$ -intercept =  $-7$

b Slope =  $6$ ,  $y$ -intercept =  $9$

c Slope =  $-1$ ,  $y$ -intercept =  $0$

d Slope =  $-1$ ,  $y$ -intercept =  $-8$

3

a  
i  $y = 15x - 10$

ii Slope =  $15$ ,  $y$ -intercept =  $-10$

b  
i  $y = 4x + 3$

ii Slope =  $4$ ,  $y$ -intercept =  $3$

4

a  $-\frac{1}{9}$

b  $-\frac{1}{4}$

5

a  $y = -8x - 9$

b  $y = -5x + 7$

c  $y = \frac{2x}{3} + 2$

d  $y = -6x - 7$

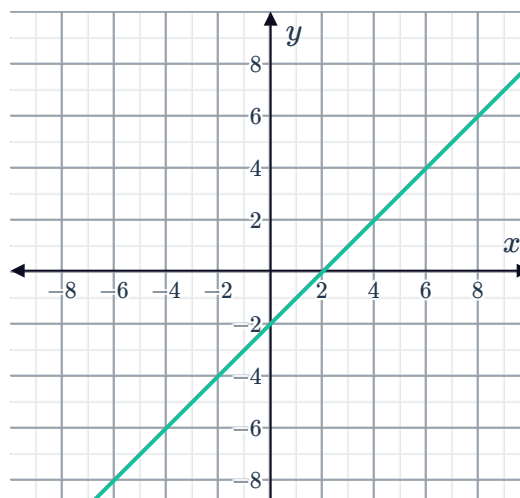
e  $y = \frac{2}{13}$

6

a  $9$

b  $y = 9x - 1$

7





8

a

i  $\frac{2}{5}$

ii 2

iii  $y = \frac{2x}{5} + 2$

b

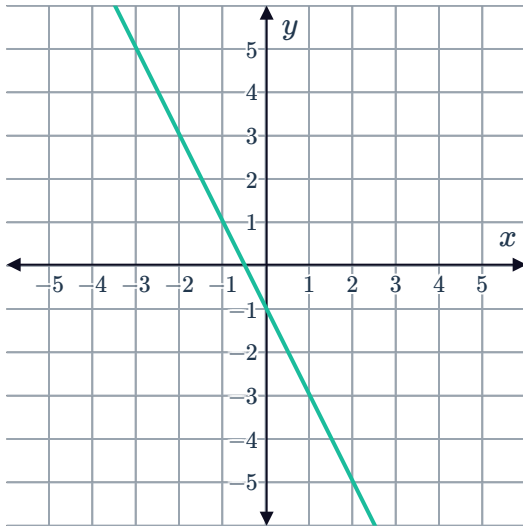
i 3

ii -1

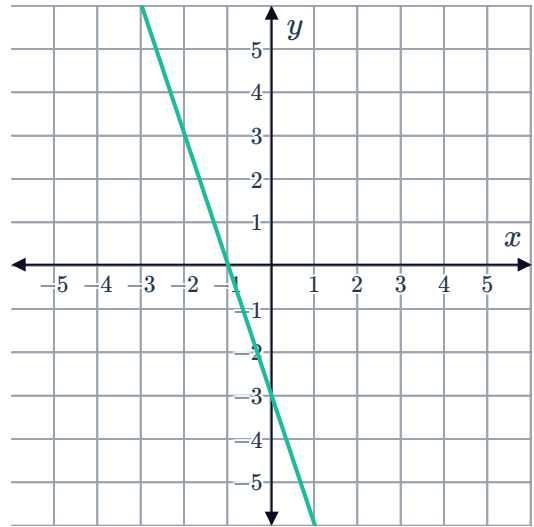
iii  $y = 3x - 1$

9

a



b



10

a  $b = -23$

b  $y = -3x - 23$

11

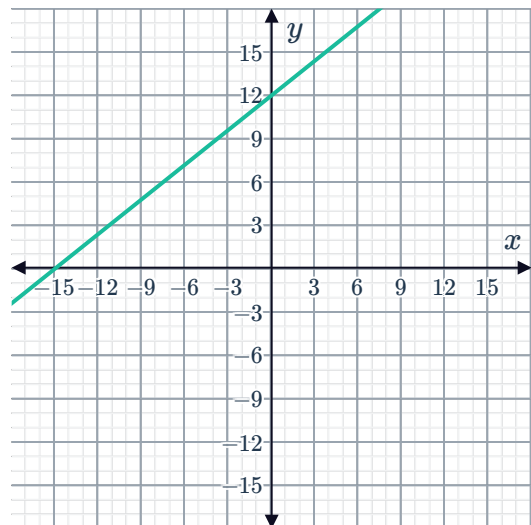
a

i  $b = 12$

iii  $(-15, 0)$

ii  $y = \frac{4x}{5} + 12$

iv

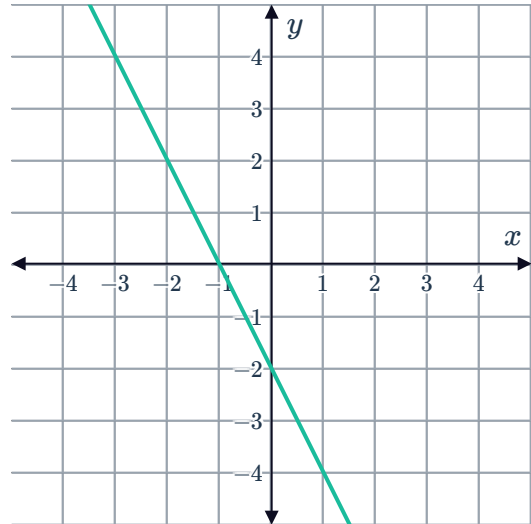


- b**
- i  $b = -2$
  - iii  $(-1, 0)$



**ii**  $y = -2x - 2$

**iv**



**12 a**  $y = -x + 2$

**b**  $a = 8$

**13 a**  $y = 1 + 8x$

**b**

|     |    |   |   |
|-----|----|---|---|
| $x$ | 0  | 1 | 2 |
| $y$ | -3 | 2 | 7 |

**14**  $y + 6x = 2$

**15**

|     |   |   |    |
|-----|---|---|----|
| $x$ | 0 | 1 | 2  |
| $y$ | 3 | 7 | 11 |

**16** Larger

**17** Smaller

**18** More rapidly

**19 A**

20 Smaller



21 Larger

22

a 60 km/h

b 71 km/h

c Bus

23

a  $u = 20$

b Compunet

24

a 4 mg/min

b 5.9 mg/min

c In capsule form

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## Additional questions

25 Line 1:  $y = 2 + 6x$

Line 3:  $y = 6x$

Line 6:  $y = 6x - 2$

26

a Yes, (0, 2)

b Yes, (0, 0)

c Yes, (0, 1)

d No

27

a Yes

b No

c Yes

d No

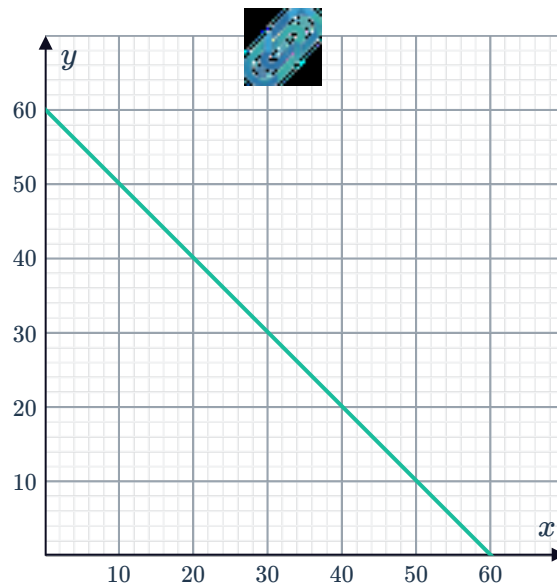
28

a

|                                     |    |    |    |    |    |    |
|-------------------------------------|----|----|----|----|----|----|
| Number of minutes passed ( $x$ )    | 0  | 5  | 10 | 15 | 20 | 60 |
| Amount of fuel left in tank ( $y$ ) | 60 | 55 | 50 | 45 | 40 | 0  |

b  $y = 60 - x$

c



29  $y = 20x + 110$

30

a  $y = 30 - 5x$

- b The race is 30 miles and he is running at 5 miles per hour, so he will finish the race in 6 hours. Time cannot be negative and he will not be running any more after 6 hours, so the restriction is  $0 \leq x \leq 6$ .

31

a  $y = -2.1x + 63$

- b The water level is falling at a constant rate of 2.1 feet per minute.
- c The water level was 63 feet before any water was pumped out.
- d 25.2 ft
- e An appropriate domain is  $x \in [0, 30]$  because solving the equation  $0 = -2.1x + 63$  we get  $x = 30$  when the well is emptied. Once the well is emptied, the context is no longer valid. Time is always positive, so we get  $[0, 30]$ .

32

a  $y = 35x + 250$

- b The total amount charged increases by \$35 for each additional hour of work.
- c The minimum amount charged by the carpenter, which is the initial fee of \$250.
- d The total cost is Materials+Labor. It will cost  $\$1800 + 250 + 35 \times 12 = 2470$ , and  $2470 < 2500$ , so she has budgeted enough for the ramp.

33



- a 1.63 ft/min
- b  $y = 1.63x$
- c 9.78 ft
- d  $x = 7$
- e The domain will be  $x \in [0, 45]$  as after 45 minutes the diver will need to stop going deeper and start coming back to the surface.
- f  $y = -1.63(x - 45) + 73.35 = -1.63x + 146.7$  for  $x \in [45, 90]$

